Oracle SQLcl Users Guide

Date: 2026-10-30 08:15:23
Document: oracle-sqlcl-users-guide_26_03.pdf
Version: Release 26.3 | G62131-02 | September 2026
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Environment:

- Directory = H:\20_projects\FJ12a_HMX-dev\44_project!about_44_project.txt
-

Project command

Create project

```
SQLcl> project init -name hmol_hmx -schemas hmol -d /workspace/ -con tdq02cei7gikc7j5_medium
or
SQLcl> project init -name hmol_ws02 -schemas hmol -con tdq02cei7gikc7j5_medium
```

Generates project tree:

```
    H:.
    +---workspace
        +---.dbtools
        ¦   +---filters
        +---dist
        +---src
            +---database
                +---admin
                ¦   +---object_grants
                +---hmol
                ¦   +---comments
                ¦   +---object_grants
                ¦   +---package_bodies
                ¦   +---package_specs
                ¦   +---procedures
                ¦   +---ref_constraints
                ¦   +---sequences
                ¦   +---synonyms
                ¦   +---tables
                ¦   +---triggers
                ¦   +---type_bodies
                ¦   +---type_specs
                ¦   +---views
                +---ut3
                    +---object_grants
```

Interesting

Project "home" (~\ ) = H:\20_projects\FJ12a_HMX-dev\44_project_

- ~\workspace.dbtools\project.config.json
  configuration for the project
- ~\workspace.dbtools\project.sqlformat.xml
  SQL formatting rules

Step into the project

```
SQLcl> cd workspace
```

Export the database contents (of one schema?)

```
SQLcl> project export -schemas hmol
```

... has exported the whole HMOL schema plus the grants to HMOL in other schema's.
Probably the filters what to export is in ~.dbtools\filters\ddl.filters

```
SQLcl> proj cfg -li
```

Result:

SETTING NAME	VALUE
project	hmol_hmx
sqlcl.connectionName	tdq02cei7gikc7j5_medium
sqlcl.autoConnect	false
sqlcl.version	26.3.0.0
schemas	["HMOL"]
lastReleaseVersion	
apex.apexlang	false
dml	false
export.format.enable	true
export.setTransform.collationClause	NEVER
export.setTransform.constraints	true
export.setTransform.constraintsAsAlter	true
export.setTransform.emitSchema	true
export.setTransform.force	true
export.setTransform.partitioning	false
export.setTransform.segmentAttributes	false
export.setTransform.sizeByteKeyword	true
export.setTransform.sqlterminator	true
export.setTransform.storage	false
export.setTransform.tablespace	false
export.apex.exptype	["APEXLANG","APPLICATION_SOURCE"]
export.apex.expAclAssignments	false
export.apex.expComments	false
export.apex.skipExportDate	true

export.apex.expirNotif	false
export.apex.expOriginalIds	false
export.apex.expSavedReports	false
export.apex.expPubReports	true
export.apex.expSubscriptions	false
export.apex.expSupportingObjects	N
export.apex.expTranslations	false
export.apex.auditType	NULL
git.protectedBranches	["main","master"]
git.defaultBranch	main
stage.excludeObjects	["ALL.user"]
stage.generatedFormat	liquibase
stage.softObjectIsolation	change
stage.substituteSchemas	false

Example workflow git & SQLcl

Line prefixes used: remote = GitHub local = on your PC or laptop SQLcl = local in a SQLcl session DB = in the database e.g. using Oracle SQL Developer or VScode with the SqlDev extension. # = comments

Steps:

1. remote: Create a new repository, e.g.: FJ30a_CICD
2. local : cd to where you want to work
3. local : mkdir ws03
4. local : cd ws03
5. local : git clone git@github.com:HermanMol/FJ30a_CICD.git .
6. local : start SQLcl: oci1 hmol
7. SQLcl : project init -name hmol_ws03 -schemas hmol -con tdq02cei7gikc7j5_medium
8. SQLcl : project export -schemas hmol
9. SQLcl : exit
10. local : git pull
11. local : git add --all
12. local : git commit -m "initial_ws03"
13. local : git push
    # Local, DB and remote are now on branch main and are in sync.
14. remote: Create new branch, e.g.: *ws03-b001*
15. local : git pull
    # New branch *ws03-b001* is now on local
16. local : git checkout *ws03-b001*
    # Now on the new branch
17. DB : Make a change to an object.
    # For this example changing package spec HMX_ANNOTATIONS
18. local : Be shure you are in the ws03 folder where you cloned the repo en exported the DB project
19. SQLcl : project export -schemas hmol
    # Exports all from the DB; changes source on local
20. SQLcl : !git add --all
21. SQLcl : !git commit -m "Changes 30-10-2026 for testing SQLcl project with git on OCI"
22. SQLcl : project stage
    # New files:

    # - ~\ws03\dist
    # - ~\ws03\dist\env
    # - ~\ws03\dist\env\default.properties
    # - ~\ws03\dist\install.sql
    # - ~\ws03\dist\releases
    # - ~\ws03\dist\releases\main.changelog.xml
    # - ~\ws03\dist\releases\next
    # - ~\ws03\dist\releases\next\changes
    # - ~\ws03\dist\releases\next\changes\ws03-b001
    # - ~\ws03\dist\releases\next\changes\ws03-b001\hmol
    # - ~\ws03\dist\releases\next\changes\ws03-b001\hmol\package_specs
    # - ~\ws03\dist\releases\next\changes\ws03-b001\hmol\package_specs\hmx_annotations.sql
    # - ~\ws03\dist\releases\next\changes\ws03-b001\stage.changelog.xml
    # - ~\ws03\dist\releases\next\release.changelog.xml
    # - ~\ws03\dist\utils
    # - ~\ws03\dist\utils\prechecks.sql
    # - ~\ws03\dist\utils\recompile.sql
23. SQLcl : exit
24. local : git status
    # Shows the changes due to the `project stage` command in SQLcl
25. local : git commit -m "WS03 added files after the SQLcl project stage command"
26. local : git push
    # The changes are pushed to the remote repo in branch *ws03-b001*.
27. local : git checkout main
28. local : git merge ws03-b001
29. local : git commit -m "Merged branch ws03-b001"
30. local : git branch -d ws03-b001
    # Maybe later, in the remote repo?
31. local : git push